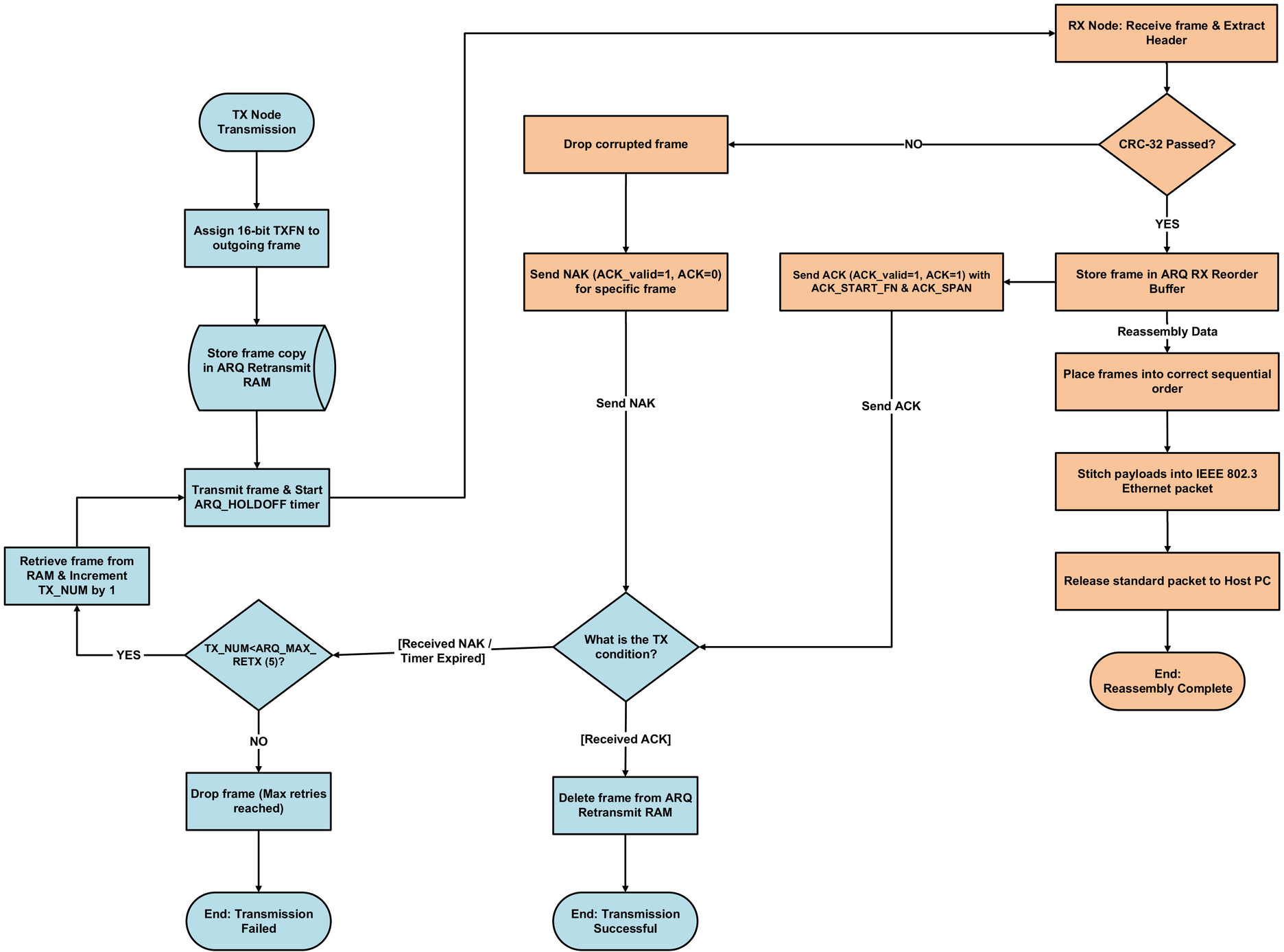

TX Node Transmission
Assign 16-bit TXFN to outgoing frame
Store frame copy in ARQ Retransmit RAM
Transmit frame & Start ARQ_HOLDOFF timer
RX Node: Receive frame & Extract Header
CRC-32 Passed?
NO
Drop corrupted frame
YES
Store frame in ARQ RX Reorder Buffer
Send ACK (ACK_valid=1, ACK=1) with ACK_START_FN & ACK_SPAN
Send NAK (ACK_valid=1, ACK=0) for specific frame
Reassembly Data
Place frames into correct sequential order
Stitch payloads into IEEE 802.3 Ethernet packet
Release standard packet to Host PC
End: Reassembly Complete
Send ACK
Send NAK
What is the TX condition?
[Received NAK / Timer Expired]
TX_NUM<ARQ_MAX_RETX (5)?
YES
Retrieve frame from RAM & Increment TX_NUM by 1
NO
Drop frame (Max retries reached)
End: Transmission Failed
[Received ACK]
Delete frame from ARQ Retransmit RAM
End: Transmission Successful